



Unit 3 · Outcome 1 · Maintaining Biodiversity

The dividends nature pays:ecosystem services

KK03

VCE Environmental Science · Units 3 & 4 · Exam study summary

Key knowledge. Ecosystems as a source of renewable services for human health and well-being: provisioning, regulating, supporting and cultural services .

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.



The dot point, in VCAA's exact words

Where this sits The dot point, in VCAA's exact words ♦ Key knowledge “Ecosystems as a source of renewable services for human health and well-being: provisioning, regulating, supporting and cultural services .” Two ideas are doing the heavy lifting in that sentence. First, renewable — these services regenerate naturally as long as the ecosystem is kept functioning. Second, human health and well-being — every service is framed by how it benefits people , which is why this dot point sits so close to value systems and sustainability later in the chapter. By

Nature's invisible utility company

The mental model Nature's invisible utility company Imagine a single company that quietly runs your water treatment plant, your power station, your farm, your insurance against floods, and your local park — and never sends an invoice. That's an ecosystem. The four “departments” of that company map exactly onto the four categories of service. Carry this analogy through the module Four departments, one free utility Provisioning is the shop floor — physical goods you take home. Regulating is the maintenance crew — keeping the climate, water and pests in

Explore the services flowing out of Kestrel Ridge

The four categories Explore the services flowing out of Kestrel Ridge Pick a category to light up where it happens in the reserve. The same patch of woodland and wetland is delivering all four at once — that's the point examiners want you to grasp. provisioning 3D scene unavailable — use the cards below. Provisioning — physical goods harvested from the reserve. Provisioning The goods you can carry away Regulating Processes kept in balance Supporting The hidden foundation Cultural Non-material benefits Provisioning service Products obtained directly from

Why supporting services hold up everything else

The exam-defining insight Why supporting services hold up everything else Supporting services — soil formation, nutrient cycling and primary production (photosynthesis) — are different from the other three. You never consume them directly, they act over very long timescales, and they are the precondition for every other service. Damage the foundation and the whole structure comes down. Press the button and watch. cause & effect 3D scene unavailable — see the explanation below. Supporting services (base) hold up provisioning, regulating and cultural. △ De

What if you had to pay for it?

The hook that makes it stick What if you had to pay for it? Because ecosystem services arrive free, we under-value them — until they're gone and we have to replace them with technology. These real cases show how expensive “free” really is, and they make brilliant exam evidence. Service category Free version The replacement bill Regulating Forested watershed naturally purifies drinking water New York City protected its Catskills watershed for ~US\$1.5 billion rather than build a filtration plant estimated at US\$6–8 billion plus running costs. Regulating Wi

Service category	Free version	The replacement bill
Regulating	Forested watershed naturally purifies drinking water	New York City protected its Catskills watershed for ~US\$1.5 billion rather than build a filtration plant estimated at US\$6–8 billion plus running costs.
Regulating	Wild bees and insects pollinate crops	In parts of Sichuan, China, collapsed pollinator populations force farmers to pollinate pear blossom by hand with paintbrushes — slow, costly human labour.
Supporting	Decomposers and weathering build fertile soil	A few centimetres of topsoil take centuries to form. Once eroded, restoring productivity means years of fertiliser, irrigation and replanting .
Cultural	Reserves provide recreation & mental restoration	Hard to price — but nature-based tourism and the documented mental-health value of green space run to billions annually , and can't simply be rebuilt elsewhere.

Sort the service into the right category

◇ ACTIVE RECALL

Practise the skill examiners test Sort the service into the right category The single most common ecosystem-services question is “classify this example.” Drag each example into a department of nature’s utility company — or tap it, then tap a bin. Get instant feedback. Provisioning drop goods here Regulating drop processes here Supporting drop foundations here Cultural drop experiences here 0 / 12 placed Tip: if you’re unsure, ask “is it a thing, a process, a foundation, or an experience?” ↻

Command-word decoder

◆ COMMAND WORDS

Read the question properly Command-word decoder Marks are lost when students describe a question that asked them to evaluate . Tap a command word to see exactly what it’s demanding of you in an ecosystem-services answer. ◆ Identify

The P·E·L writing trainer

P·E·L WRITING

Structure that scores The P·E·L writing trainer High-mark ecosystem-services answers follow a rhythm: make a Point , back it with Evidence (a named example), then Link it to human well-being or the question. Watch one well-built sentence. P · POINT State the service & category Name the specific service and which of the four categories it belongs to. Don't be vague — “provides things” earns nothing. E · EVIDENCE Give a concrete example Anchor it in a real or case-study example. A named species, place or mechanism shows you understand, not just memorise. L



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Five questions — try, then reveal model answers

★ PRACTICE & SELF-MARK

Exam practice · 22 marks Five questions — try, then reveal model answers Attempt each one in your logbook first. Then reveal a weak and a strong response, the marking guide, and award yourself a mark. Your running score sits in the dock, bottom-right.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.